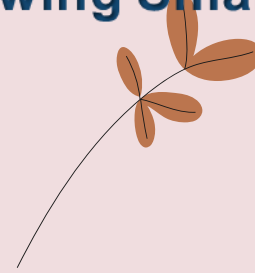




# Medical Assistive Robot

Human-Following Smart Dustbin.



**Project title: Human-Following Smart Dustbin**

**Course Name: Robotics Lab**

**Course Code: CSE 438**



**Presented by:**

Name: Jannatul Maowa Meghla

Student ID: 22201194

**Groupmates:**

Name: Hridita Laskar

Student ID: 22201184

Name: Meher Afroz

Student ID: 22201176



**Submitted To:**

**Dr. Nazmun Nahid**

Assistant Professor

Department of Computer Science and Engineering,  
University of Asia Pacific



# Executive Summary

01

**Medical Assistive Robot is a smart autonomous robot designed for hospital environments.**

02

**The robot automatically follows nurses during patient visits.**

03

**A tray system is included to carry important medical tools and supplies.**

04

**Includes automatic waste disposal**

05

**Helps reduce infection & contamination risks.**

06

**The system reduces nurses' workload and improves efficiency.**



08

**IR and ultrasonic sensors ensure safe movement and distance control. Stops safely near patients or obstacles**

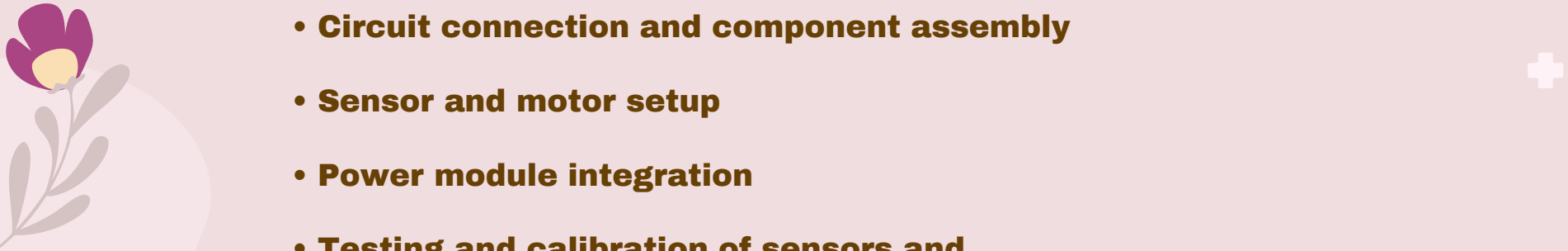
09

**The project demonstrates The project demonstrates robotics applications in smart healthcare assistance.**

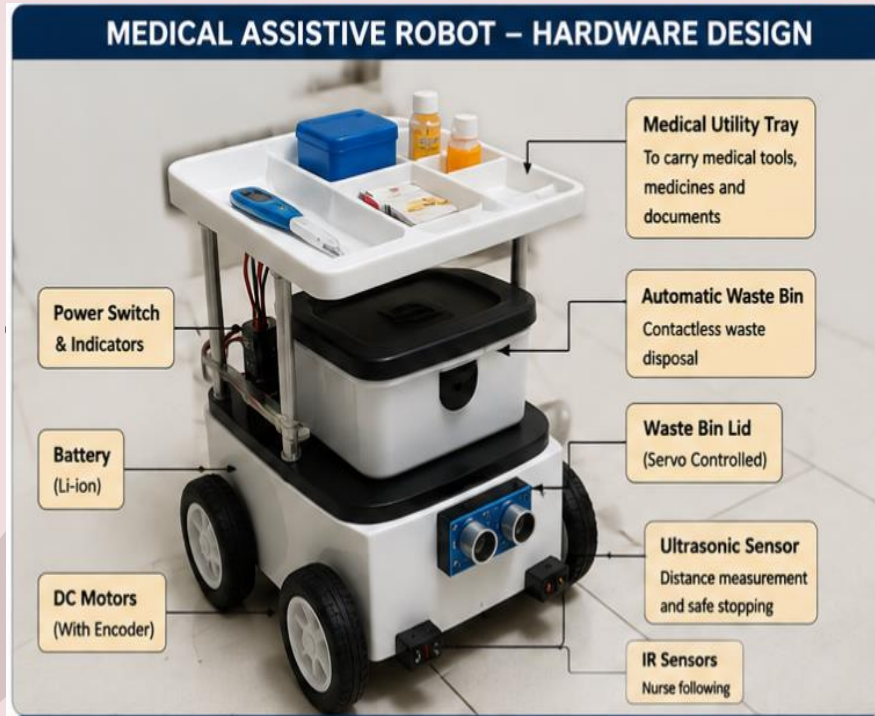




# My Contribution (Hardware)

- 
- **Hardware design and integration**
  - **Circuit connection and component assembly**
  - **Sensor and motor setup**
  - **Power module integration**
  - **Testing and calibration of sensors and actuators**
  - **Chassis and embedded system arrangement**
  - **Circuit diagram preparation**
- 

# Hardware design



**Inspiration Diagram**



**Robot Diagram**

# components



**Arduino Uno**



**HC-SR04**



**IR Sensors**



**Chassis**



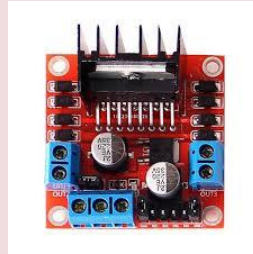
**Servo Motor**



**Battery**



**Breadboard**



**L298N Motor Driver**

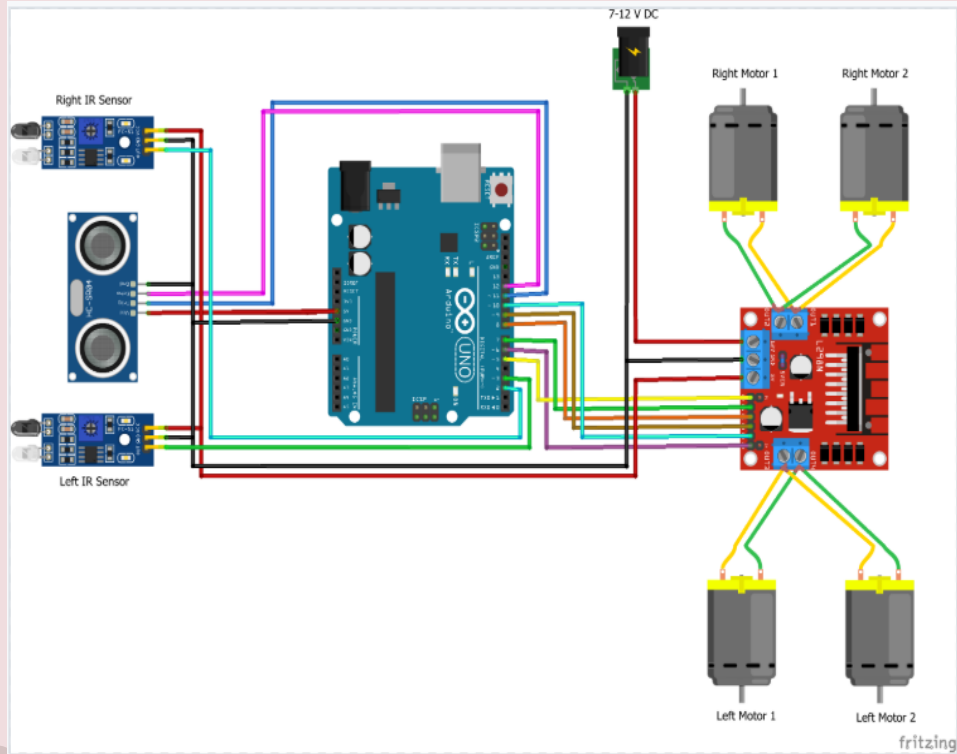


**Dc motor**

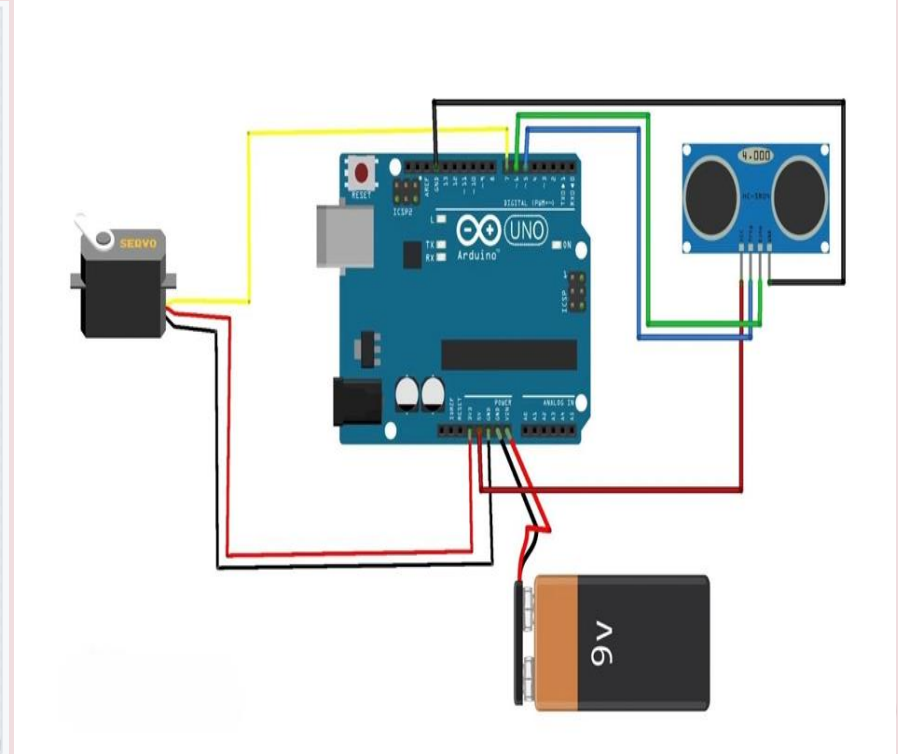


**Waste disposal bin**

# Circuit Diagram



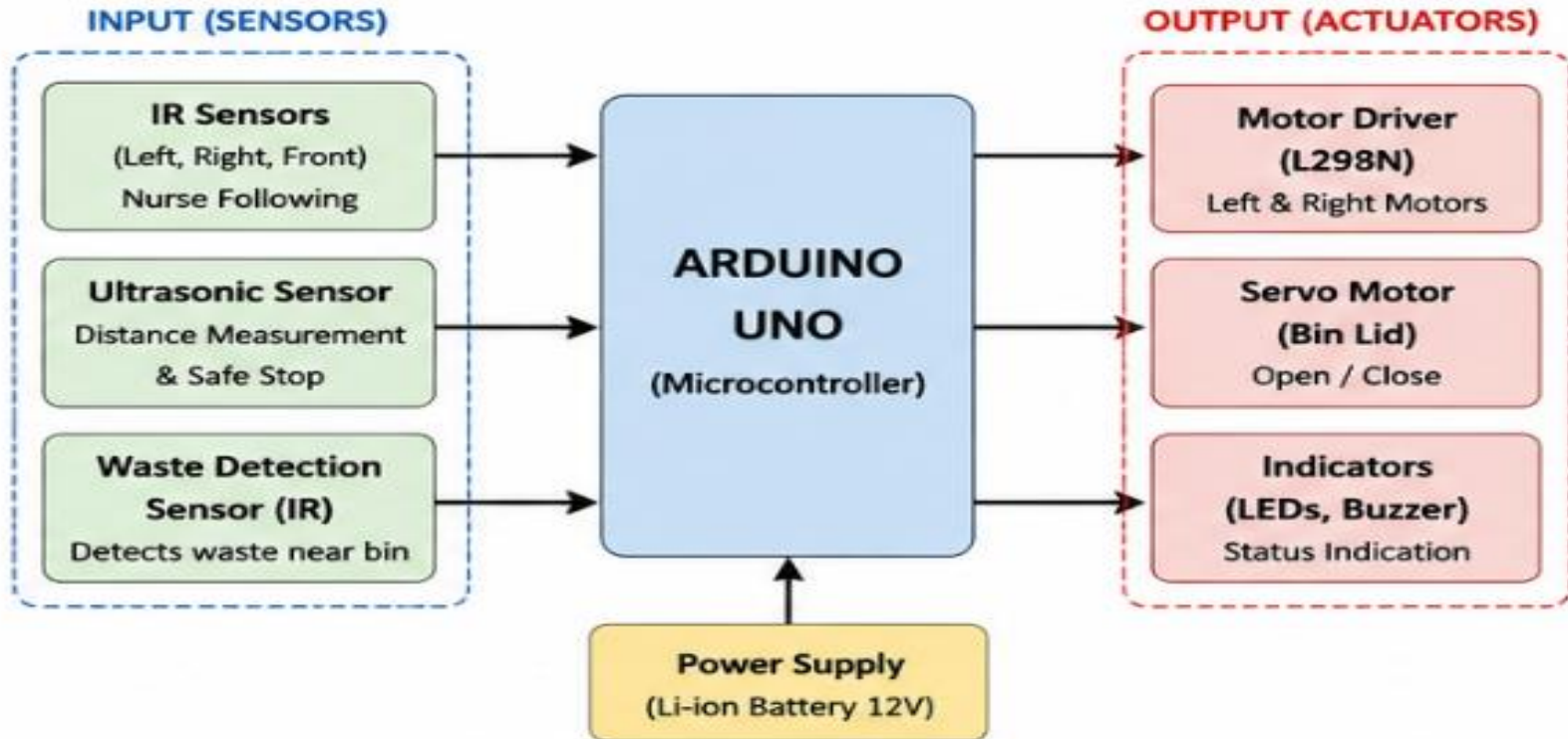
**Human following car**



**Automatic Dustbin**

# Embedded system

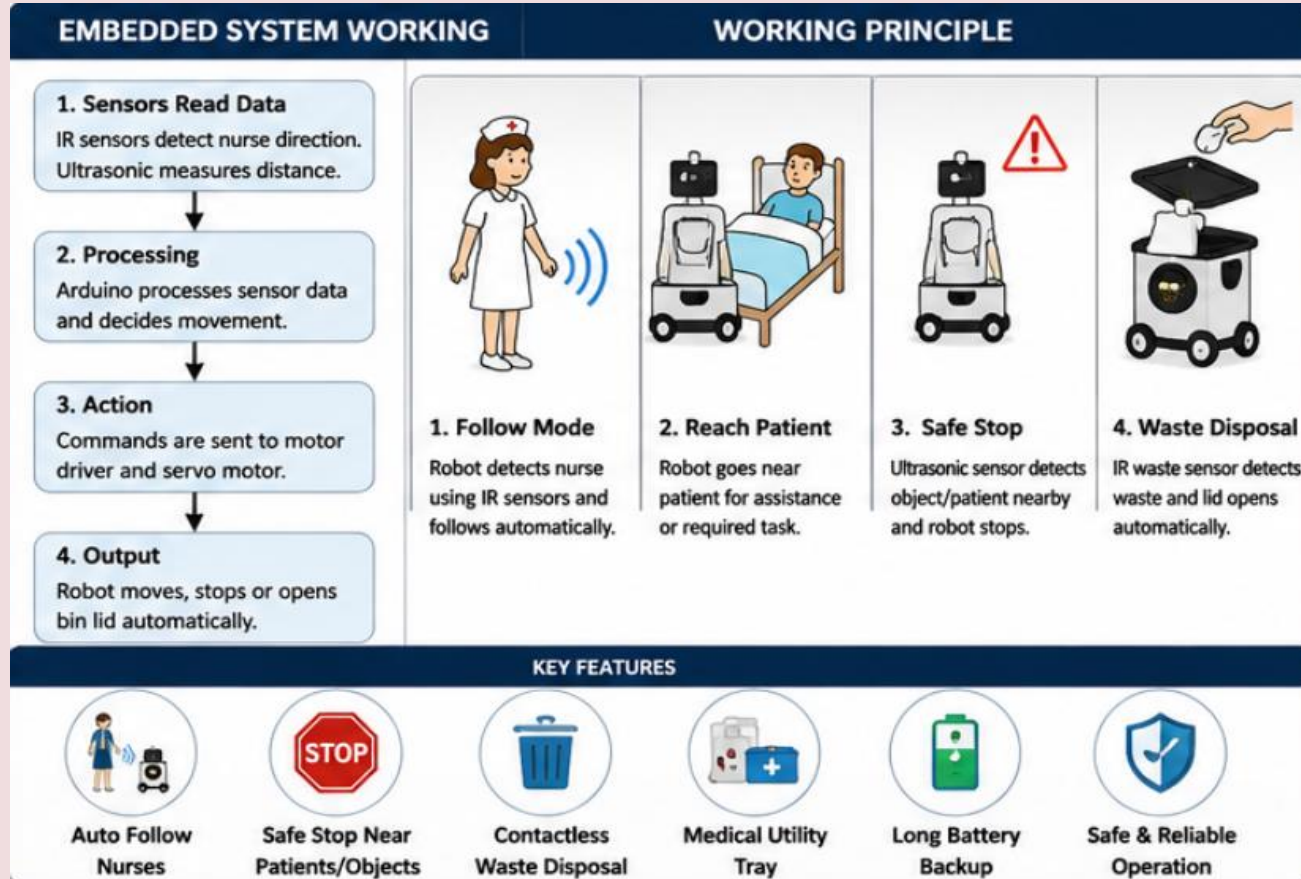
## EMBEDDED SYSTEM BLOCK DIAGRAM





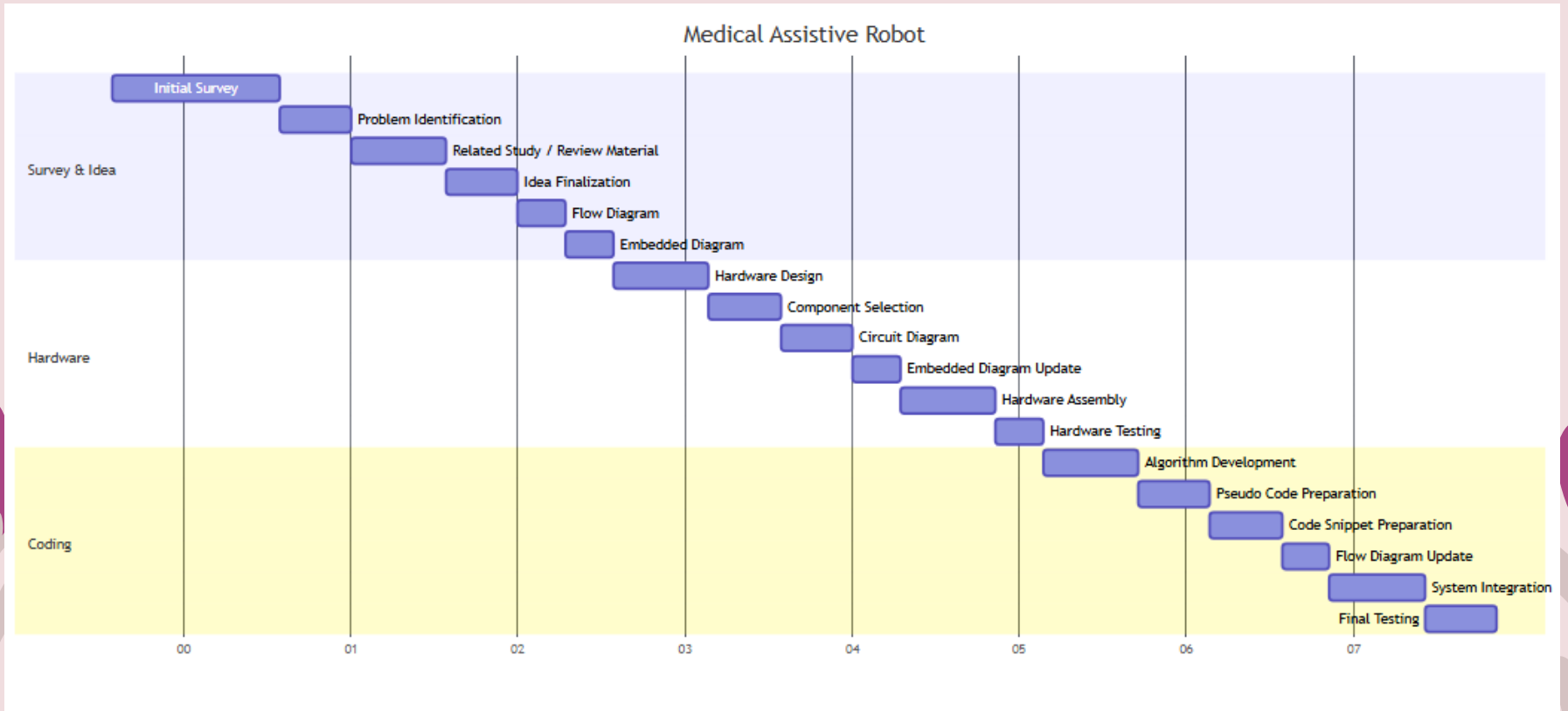


# Working principle





# Gantt chart





# Conclusion



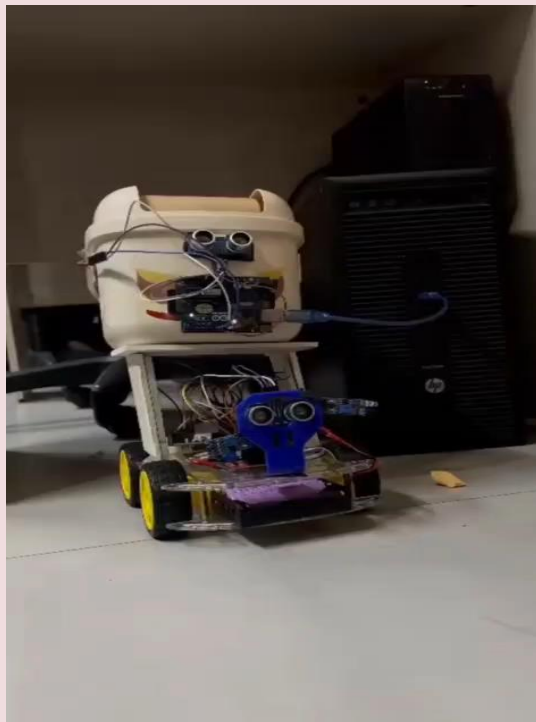
**The Medical Assistive Robot provides an efficient and hygienic solution for hospital assistance and waste management.**



**The system reduces nurses' workload through automatic following and contactless waste disposal features.**



# Demo video



**Drive link:** [https://drive.google.com/file/d/1IPtGXYWmlhWZWMSduGhqsJLswSWoy-lw/view?usp=drive\\_link](https://drive.google.com/file/d/1IPtGXYWmlhWZWMSduGhqsJLswSWoy-lw/view?usp=drive_link)

The background is a light pink color with various decorative elements. On the left side, there are three purple flowers with yellow centers and grey stems. On the right side, there are brown leaves and grey wavy shapes. In the center, there is a large, light pink, irregular shape. Scattered around are small icons: a white cross, a small circle, a triangle, a pill, and a small circle with a plus sign.

Any  
question??

**Thank you**

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